

David O'Neil

NLP / LLM Engineer · Technical Lead

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NLP/LLM engineer with 4+ years. Technical lead for training CEMIG's in-house LLMs (CEIA/UFG) and NLP Team Lead on Digital Humans II (AKCIT/EMBRAPII). Accepted contributor to official NVIDIA repositories (NeMo, Megatron, Nemotron). Papers at ICLR, ICML, and ACL. Production NLP at TCE-GO and TCM-GO — hybrid RAG, Graph RAG, MLOps; reported R\$1.8M in savings at TCM-GO.

EXPERIENCE

NLP Team Lead · Research

Digital Humans II — AKCIT / EMBRAPII

- Led the NLP team (direct reports). Mandate: base research and publications — interpretability, persona alignment, episodic memory — not product engineering.
- Research on Sparse Autoencoders, steering, Steering-Conditioned LoRA, persona alignment, episodic memory, and inference acceleration.
- Three accepted workshop papers (ICLR MemAgents oral, ICML CoLoRAI, ACL SemEval); two manuscripts under review at top-tier venues.

Technical Lead · Model Training

CEMIG — CEIA / UFG

- Technical lead for CEMIG's in-house LLMs: synthetic-data strategy, internal data use, and eval (public benchmarks vs. domain criteria such as ANEEL factual accuracy).
- Ran SFT, LoRA, full fine-tuning, DPO, and GRPO on Qwen-3/3.5, Gemma-4, LLaMA 3.1/3.2, and Tucano on NVIDIA Blackwell B200, DGX H100, and A100 (NeMo, Megatron, Axolotl, Unsloth).
- Accepted commits/PRs in official NVIDIA repositories (NeMo / Megatron / Nemotron).

Tech Lead & NLP Engineer

TRON — CEIA

- Tech lead for CEIAxTRON copilot: Graph RAG with specialized agents (query rewrite, rerank) and actions inside a legacy Delphi app, plus a web extension for human agents.
- Supervised fine-tuning of BGE-M3 with triplet loss and hard-negative mining to raise retrieval quality.

NLP Engineer

TCE-GO — Court of Accounts of Goiás

- Hybrid RAG (lexical + dense) over complementary legal documents; regex/metadata routing when semantic search was insufficient.
- Encoder SFT (E5, BGE-M3, Qwen Embedding) with contrastive/triplet loss; Graph RAG and agent orchestration; production chat, summarization, and QA including deploy.

NLP Engineer

TCM-GO — Municipal Court of Accounts of Goiás

- NLP chat product with reported R\$1.8M in operational savings; legal RAG and fine-tuning for document sections.
- Medallion data architecture, MLOps on institutional infra, encoder training/eval, and agent pipelines over TCM databases.

CEO & Co-Founder

Pergunt.ai

- Edtech GenAI: original public-exam questions and a recommendation layer on student performance.

2023

EDUCATION

B.S. in Artificial Intelligence

Federal University of Goiás (UFG)

First AI undergraduate program in Latin America. In progress.

2023 — 2027

PUBLICATIONS

Episodic Memory from Compression Boundaries in Latent Representation Space

ICLR 2026 · MemAgents Workshop (Oral)

Steering-Conditioned LoRA via Sparse Autoencoders

ICML 2026 · CoLoRAI Workshop (Poster)

AKCIT-UFG at SemEval-2026 Task 8: Structured Chunking and Optimized Query Reformulation for Efficient Multi-Turn Retrieval

ACL 2026 · SemEval Workshop

SKILLS

LLM post-training: SFT · LoRA / full FT · DPO / GRPO · Synthetic data · Public + custom eval · Qwen-3/3.5 · Gemma-4 · LLaMA 3.1/3.2 · Tucano · Axolotl / Unsloth

Research / interpretability: Sparse Autoencoders · Steering · Steering-Conditioned LoRA · Persona alignment · Episodic memory · Inference acceleration

Retrieval / production: E5 / BGE-M3 / Qwen Embedding · Triplet / contrastive · Graph RAG · Agent orchestration · BM25 + metadata · Medallion · MLOps

NVIDIA / compute: Blackwell B200 · DGX H100 · A100 · NeMo · Megatron · Nemotron